ADMIT-LIPA:
A WEB-BASED ADMISSIONS CUSTOMER RELATIONSHIP MANAGEMENT SYSTEM FOR THE ADMISSIONS OFFICE OF STI COLLEGE LIPA

STI College Lipa
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School of Information and Communications Technology

Prototype Progress Report

Presented in Partial Fulfillment of the Requirements in
Application Development and Emerging Technologies

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Summary of Progress

The proponents moved ADMIT-Lipa from the approved System Concept Proposal to a working web application for the Admissions Office of STI College Lipa. The system replaces the current practice in which applicants send a screenshot of their proof of payment to the campus Facebook Messenger account for validation (STI Education Services Group, n.d.). All fourteen use cases in the use-case diagram of the proposal now run end to end in the development build, and the automated acceptance test suite passed 52 of 52 checks on October 28, 2026. Each functional requirement, FR-01 to FR-14, is exercised by at least one check that follows the verification method written in the proposal. Table 1 summarizes the status of the prototype.

Table 1
Prototype Status as of October 28, 2026

Item	Status
Use cases implemented	14 of 14 (UC1 to UC14)
Functional requirements verified by automated test	14 of 14 (FR-01 to FR-14)
Non-functional requirements verified	NFR-02 and NFR-03 (in part) by automated test; NFR-05 checked at a 390-pixel viewport, with the 360-pixel check stated in the proposal still pending
Screens designed and captured	13 screens in desktop (1920 × 1080) and mobile (390 × 844) sizes
Source code	https://github.com/Tiredicey/APPDEV
Deployment	Runs in the development environment; public hosting on Cloudflare Pages is the next step

Technology Used

The proponents built the prototype on the technologies listed in Table 2. Each choice keeps hosting costs low for the campus and protects the personal data that applicants submit.

Table 2
Technology Stack of the Prototype

Layer	Choice	Reason
Server	Hono web framework on Cloudflare Workers and Pages	Runs at the network edge; no server for the campus to maintain
Database	Cloudflare D1 (SQLite)	Relational tables for tickets, requirements, payments, and the audit log; D1 encrypts data at rest and in transit (Cloudflare, 2026)
Field encryption	AES-256-GCM through the Web Crypto API	Date of birth, mobile number, address, and financial aid answers are encrypted before they reach the database (NFR-03)
Passwords	PBKDF2-SHA-256, 100,000 iterations, per-user salt	No plain-text passwords are stored
Front end	Plain JavaScript single-page application	Small download for mobile users on slow connections
Development assistant	Genspark AI coding agent	Generated code, tests, and the background video under the proponents' direction; the team reviewed the screens and tested the prototype

Requirement Traceability

Table 3 lists each functional requirement, the use case it traces to, the behavior implemented in the prototype, and the result of its automated verification.

Table 3
Functional Requirements Implemented and Verified

Req.	Use Case	Implemented Behavior	Result
FR-01	UC1	The inquiry form returns a ticket number such as ADM-2026-00001 in the same session and stores it	Pass
FR-02	UC2	A second profile with the same surname, first name, and date of birth is blocked with a message; the check works on encrypted data through a keyed index	Pass
FR-03	UC3	Each required document shows received, pending, or rejected; a rejection requires a reason	Pass

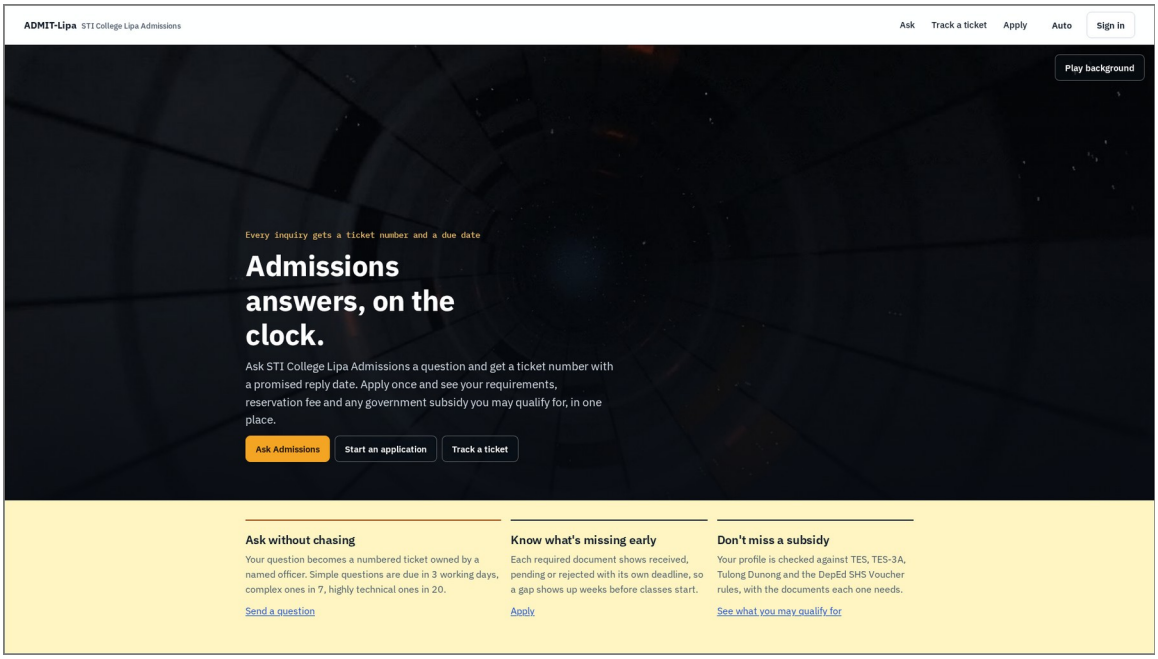
Req.	Use Case	Implemented Behavior	Result
FR-04	UC4	TES, TES-3A, and TDP (Unified Student Financial Assistance System for Tertiary Education, n.d.) and the SHS Voucher (Department of Education, n.d.) are flagged from the encoded profile; the income ceiling was tested at P400,000 and P400,001	Pass
FR-05	UC5	An applicant sees only his or her own stage; opening another record is refused	Pass
FR-06	UC6	Elapsed working days are computed against the 3-, 7-, and 20-day tiers, excluding weekends and holidays entered by the administrator	Pass
FR-07	UC7	Volume, stage conversion, and case aging are reported for any date range; totals reconcile with the raw case list; the report exports to CSV	Pass
FR-08	UC8	Consent statement text, timestamp, and purpose are stored for every inquiry and application; no consent, no record	Pass
FR-09	UC9	A reservation fee is validated only with an official receipt number, and a receipt number cannot be reused	Pass
FR-10	UC10	The deficiency notice lists each document still pending after its deadline; repeat runs do not duplicate notices	Pass
FR-11	UC11	An endorsed case reaches the financial aid coordinator queue with the document list of its program	Pass
FR-12	UC12	Opening a record requires a stated purpose; the user, record, time, and purpose are logged, and applicants can download their own access log	Pass
FR-13	UC13	Each ticket is owned by one named officer and cannot be resolved without a closing remark	Pass
FR-14	UC14	An overdue ticket is raised to the School Administrator queue; on-time tickets are not	Pass

Prototype Screens

Figures 1 to 6 were captured from the running prototype with an automated browser on October 28, 2026. The landing page uses the 1921 × 1080 Landing Page frame and its #FFF5C3 fill from the team's Figma file as its base.

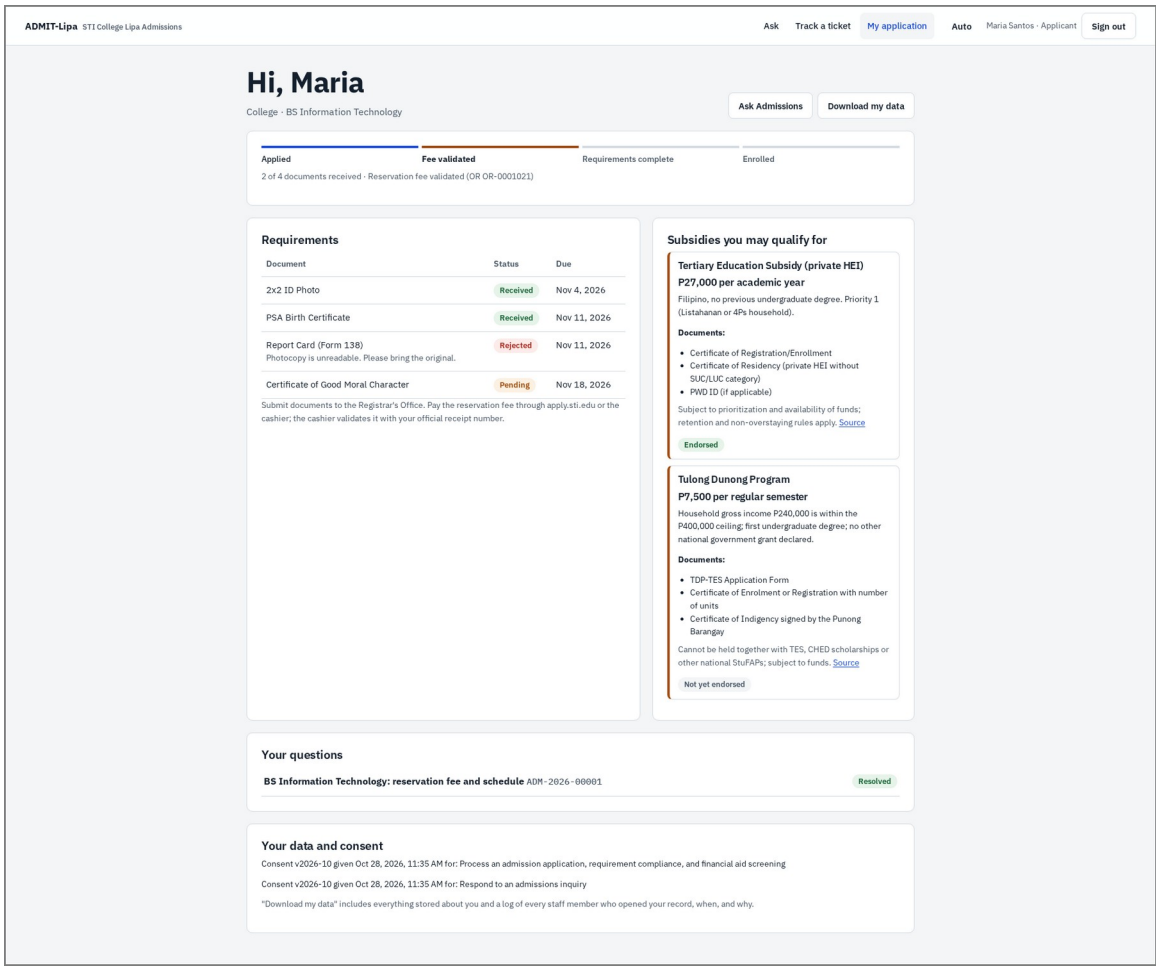
Figure 1

Landing Page With the Looping Background Video



Note. Desktop view seen by applicants and parents before signing in.

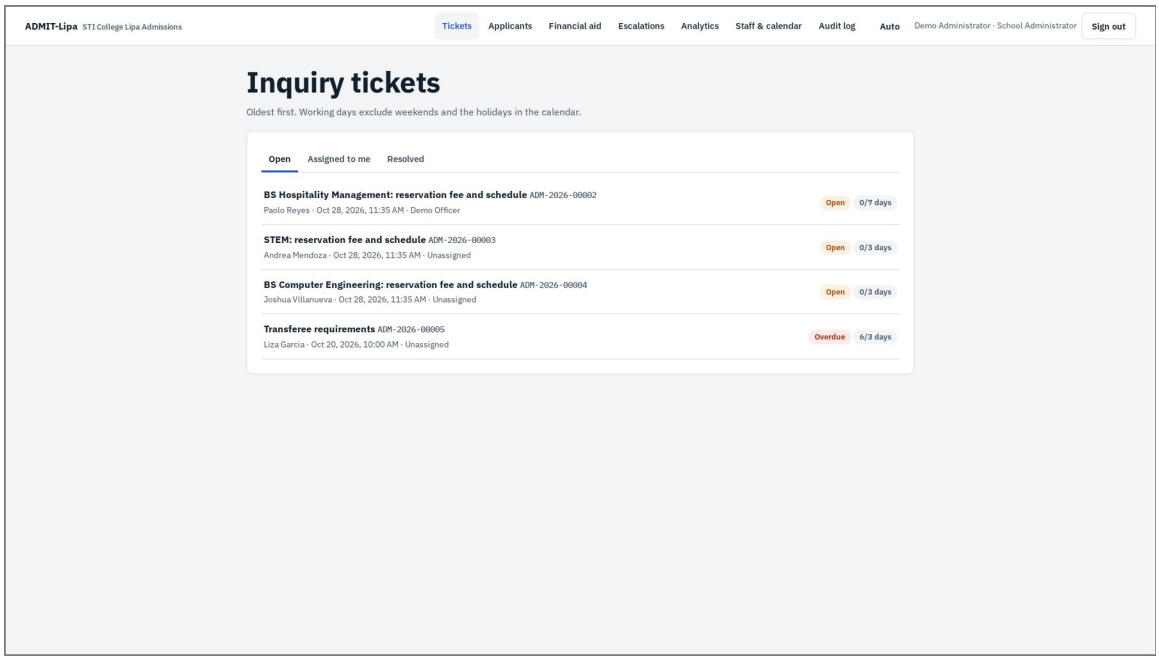
Figure 2
Applicant Dashboard



Note. Shows the application stage, requirement status, subsidy matches, inquiries, and consent records (UC3, UC4, UC5, UC8).

Figure 3

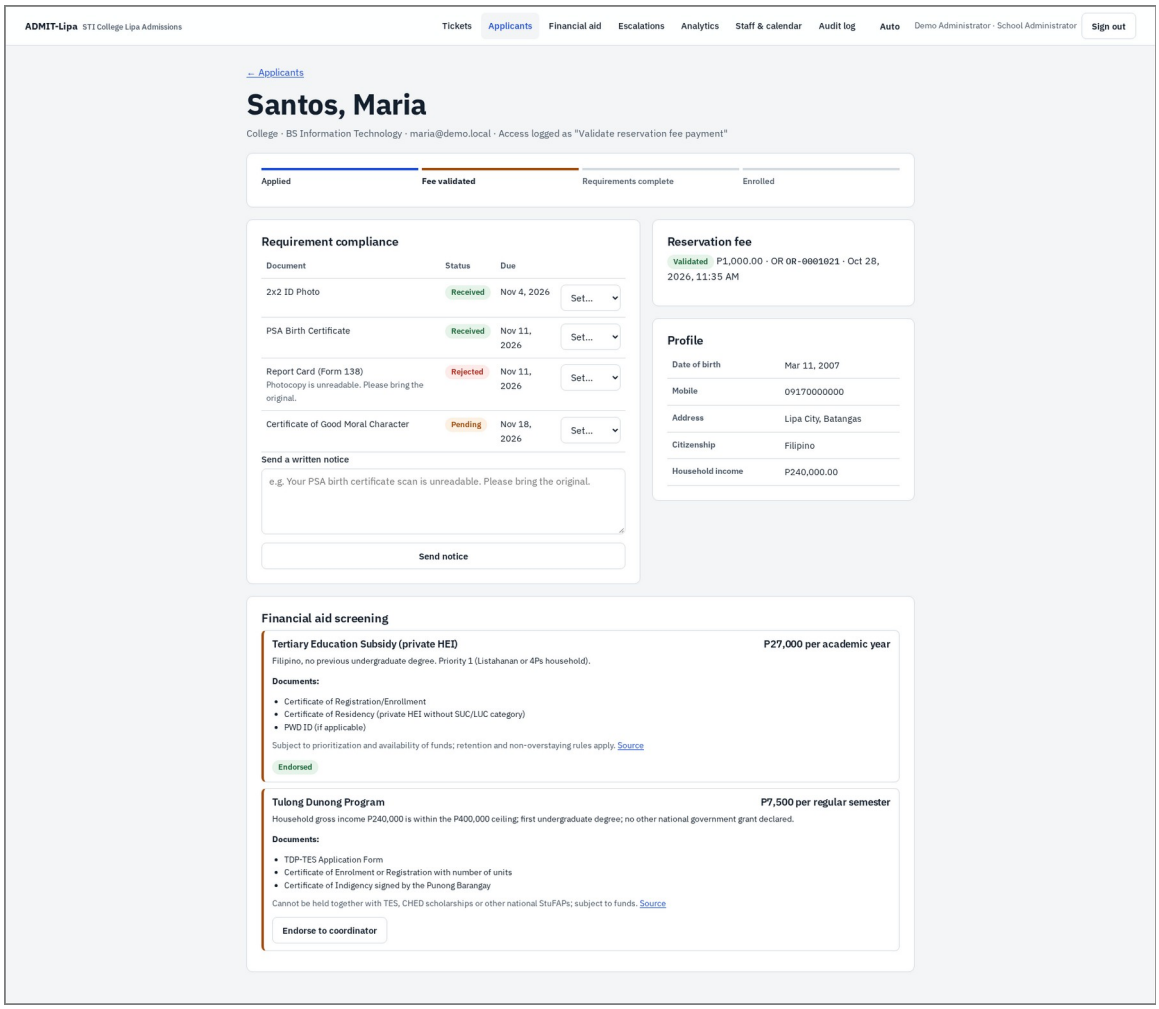
Admissions Officer Ticket Queue



Note. Each ticket shows its owner and its elapsed working days against its service tier (UC6, UC13).

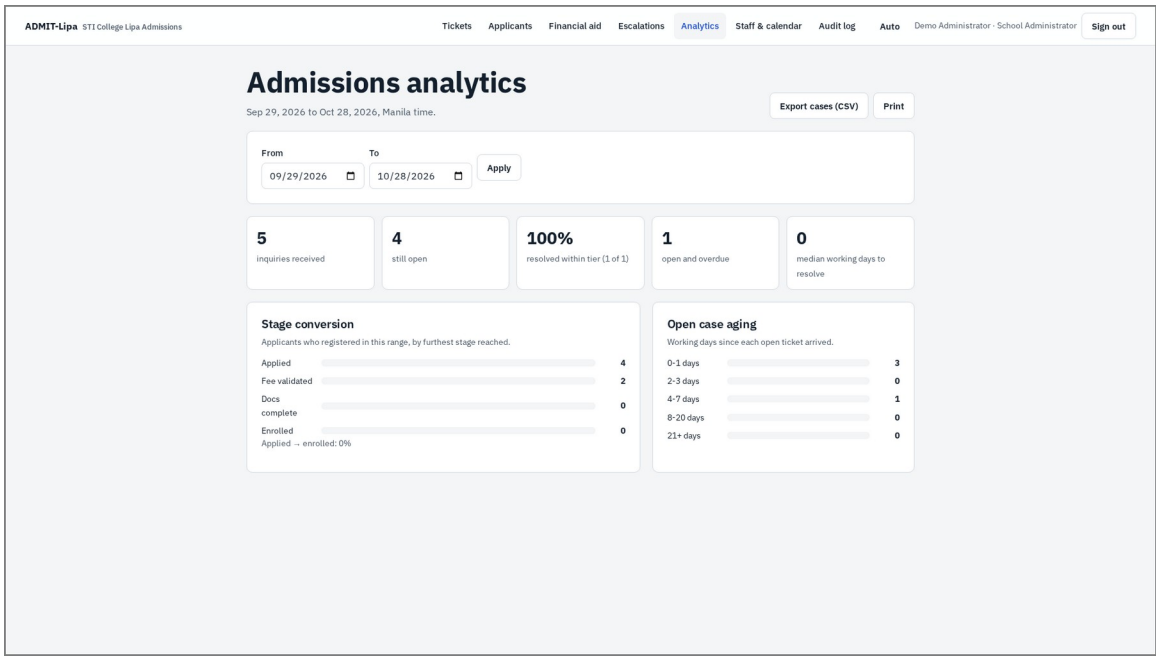
Figure 4

Staff View of an Applicant Record



Note. Covers requirement compliance, reservation fee validation, and subsidy endorsement (UC3, UC9, UC10, UC11).

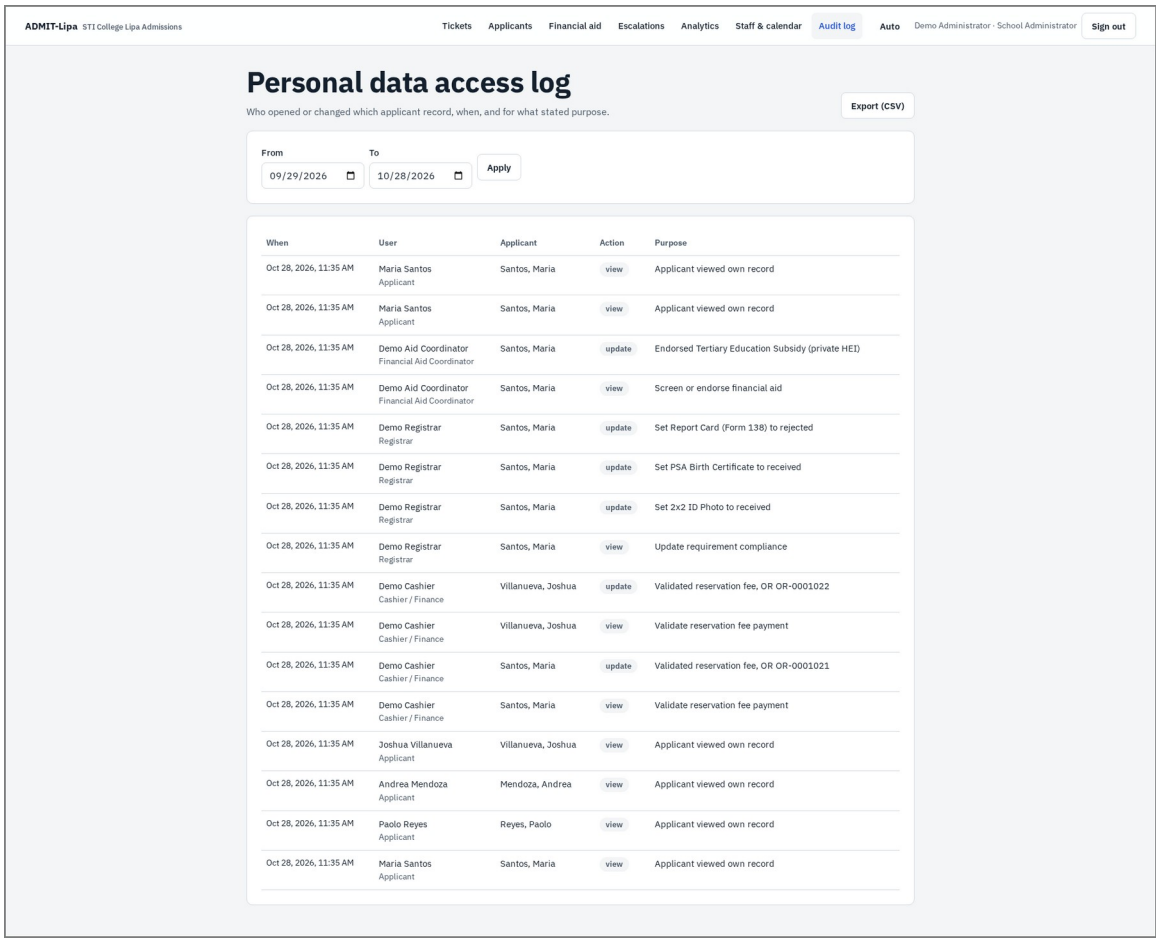
Figure 5
Admissions Analytics for the School Administrator



Note. Inquiry volume, service-level compliance, stage conversion, and case aging for a selected date range (UC7).

Figure 6

Personal Data Access Log for the Data Protection Officer



Note. Each entry records who opened which applicant record, when, and for what stated purpose (UC12).

Background Video

The landing page plays a 12-second seamless loop at 30 frames per second, 1280 × 720 pixels, 6.3 MB, encoded in H.264. It shows a slow fall down the core of an abandoned orbital station. Camera speed and the amber warning lamps pulse at 140 beats per minute. The largest frame-to-frame brightness change is 1.56 of 255 levels, far below the flash threshold of Success Criterion 2.3.1 (World Wide Web Consortium, 2023). The video is muted, carries no information, and has a Pause control. It does not play automatically for visitors who set their device to reduce motion.

Member Contributions

Table 4 lists the role and contribution of each member. Each member signs his own row to confirm the contribution listed.

Table 4
Member Roles and Contributions

Member	Role	Contribution	Signature
Atienza, Kurt Judd P.	Project Lead	Designed the landing frame in Figma; directed the AI-assisted build of the back end, database, and test suite; integrated and tested the prototype	
Garonia, Ian Humprey M.	UX/UI Designer and Tester	Designed screens in Figma; tested the prototype	
Apura, Alexis H.	UX/UI Designer and Tester	Designed screens in Figma; tested the prototype	
Tenedo, Francis Gerald G.	UX/UI Designer and Tester	Designed screens in Figma; tested the prototype	

References

Cloudflare. (2026, April 21). *Data security*. Cloudflare D1 Docs. https://developers.cloudflare.com/d1/reference/data-security/

Department of Education. (n.d.). *Senior high school voucher program guidelines*. DepEd Region XI. https://depedroxi.ph/voucher-program/

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World Wide Web Consortium. (2023). *Web content accessibility guidelines (WCAG) 2.2*. https://www.w3.org/TR/WCAG22/